

SRI CHAITANYA IIT ACADEMY, INDIA

AP • TELANGANA • KARNATAKA • TAMILNADU • MAHARASHTRA • DELHI • RANCHI

**PRAGYA G8 – ACHIEVERS | MATHEMATICS
ZERO REPORT | INTEGRATED PAPER – I (JEE MAIN)**

Paper	Integrated Paper – I, JEE Main pattern (file: Paper_I_MAIN . pdf)
Batch	PRAGYA G8 – Achievers
Syllabus	Surds; Logarithms; Sequences & Series; Quadratic Equations
Pattern	25 questions – Section A: Q1–20 single correct; Section B: Q21–25 numerical value
Date of report	02-10-2026
Result	All 25 keys verified. 7 corrections made (listed below). Outstanding errors: 0.

ANSWER KEY

1	2	3	4	5	6	7	8	9	10
(B)	(D)	(A)	(C)	(D)	(C)	(D)	(B)	(C)	(A)
11	12	13	14	15	16	17	18	19	20
(B)	(D)	(A)	(C)	(B)	(C)	(B)	(A)	(D)	(A)
21	22	23	24	25					
13	7	2	3	23					

SUMMARY

Key distribution (Section A)	(A) 5 (B) 5 (C) 5 (D) 5 – no two consecutive questions share a letter
Section B answers	all non-negative integers (13, 7, 2, 3, 23)
Difficulty	Easy 10 Medium 11 Difficult 4
Chapter usage	Surds 15 Logarithms 15 Seq. & Series 23 Quadratic Eqns 17 (questions touching each chapter)
Syllabus / method level	Within G8 CPT-01 scope; no excluded topic (AGP, sigma, V_n , telescopic series, location of roots, log inequalities, graphs); no calculus, Cauchy–Schwarz, Newton sums, cubics, trigonometry or GIF.

QUESTION-WISE ANALYSIS

Q	Type	Chapters	Concept tested	Level	Key
1	SCQ	Surds, QE, S&S	Denesting $\sqrt{a - 2\sqrt{b}}$; conjugate root; AP	M	(B)
2	SCQ	QE, S&S, Logs	$f^8 = 1$: base 1 / index 0 / base -1 with even index	M	(D)
3	SCQ	Logs, Surds, QE	Reciprocal bases: $\log_{1/a} x = -\log_a x$	M	(A)
4	SCQ	Logs, Surds, S&S	Reciprocal bases give $\text{LHS} \equiv 0$; $N = 0$	M	(C)
5	SCQ	Logs, QE, S&S	Logs in AP $\Rightarrow$ arguments in GP; domain check	E	(D)
6	SCQ	S&S, Surds, QE	Infinite GP; rationalising; conjugate root	M	(C)
7	SCQ	Logs, QE, S&S	Take logs; product of roots via sum of log roots	M	(D)
8	SCQ	QE, Surds, S&S	$\alpha^3 - \beta^3 = (\alpha - \beta)(\alpha^2 + \alpha\beta + \beta^2)$	E	(B)
9	SCQ	Logs, QE, S&S	Integer roots: eliminate k , factorise	D	(C)
10	SCQ	S&S, QE, Surds	GP ratio from AP condition; distinctness rejects $r = 1$	M	(A)
11	SCQ	Logs, QE, S&S	$t + 1/t = 5/2$; base-change reciprocal	E	(B)
12	SCQ	Surds, QE, S&S, Logs	Infinite nested radical $\Rightarrow$ quadratic	E	(D)
13	SCQ	Surds, S&S	Denesting $\sqrt{a \pm b\sqrt{c}}$	M	(A)
14	SCQ	Logs, QE	Log equation with domain rejection	E	(C)
15	SCQ	S&S, Surds	AP as $b - d, b, b + d$; GM of extremes	E	(B)
16	SCQ	Surds, S&S	Perfect squares under root $\Rightarrow u - 2 + u - 3 = 1$	D	(C)
17	SCQ	QE, S&S	Quadratic in 2^x	E	(B)
18	SCQ	Surds, S&S, Logs	GM, HM of conjugate surds	M	(A)
19	SCQ	QE, Logs, S&S	$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$	E	(D)
20	SCQ	Logs, S&S, Surds	Chain rule of logarithms	E	(A)
21	NVA	Logs, Surds, S&S	Characteristic; digits and leading zeros	M	13
22	NVA	QE, Surds, S&S	$\alpha^3 - \beta^3$ via factorisation	M	7
23	NVA	S&S, QE, Logs	$1 + r^2 + r^4 = (1 + r + r^2)(1 - r + r^2)$	D	2
24	NVA	Logs, S&S	Classical identity $ab + 5(a - b) = 1$	D	3
25	NVA	S&S, Surds, QE	AP terms; product of roots	E	23

SCQ = single correct; NVA = numerical value answer; E/M/D = easy/medium/difficult.

ERRORS FOUND AND CORRECTED

- Title / header.** “Hell Paper” and “(HELL LEVEL)” removed. Paper is now *Integrated Paper – I, JEE Main pattern*; running header reads *Paper I – Main*.
- Tagline removed.** “Every question draws on at least three of: Surds, Logarithms, Sequences & Series, Quadratic Equations” was also not true – Q13, Q14, Q15, Q16, Q17 and Q24 each use two chapters (see analysis table).
- Instruction and marking-scheme boxes removed** from both sections. The Section B box was also wrong for a numerical section (“+4 if the correct *option* is chosen”, -1 negative marking).

4. **Q19 – ambiguous notation.** $\log_2(\alpha - \beta)^2$ can be read as $[\log_2(\alpha - \beta)]^2$; rewritten as $\log_2[(\alpha - \beta)^2]$. Key unchanged.
5. **Q10 – wording.** “the sum to infinity of the geometric progression” now names it: “... of the geometric progression $a, b, c, \dots$ ”. Key unchanged.
6. **Page breaks.** Q5 and Q19 had their options printed on the next page, away from the stem. Every question is now kept on one page with its options; the long surd sum in Q1 is now set on its own line.
7. **Answer-letter clustering.** Overall 5/5/5/5, but (D) was the key for five of Q2–Q8 and for none of Q9–Q20, with five back-to-back repeats. Options swapped in Q3 (D→A), Q8 (D→B), Q12 (B→D), Q14 (A→C), Q19 (A→D), Q20 (C→A); same options, same answers, new letters.

No answer in the paper was wrong, no question had zero or two correct options, and no two options in any question were equal.

OBSERVATIONS (NO CHANGE MADE)

- Q8 and Q22 test the same step ($\alpha^3 - \beta^3 = (\alpha - \beta)(\alpha^2 + \alpha\beta + \beta^2)$) on different quadratics; Q1 and Q6 share one template (a surd taken as a root of $x^2 + px + q = 0$ with $p, q \in \mathbb{Z}$).
- Q2 needs the base = -1 (even index) case. A student who omits it gets $S = 10$, which is not a term of $3, 7, 11, \dots$ – so there is no rival answer to object with.
- Q4 rests on $N = 0$ (the two logarithms cancel identically); that is the intended trap.

KEY VERIFICATION

- Q1 (B)** $s = \sqrt{6} - 1 \Rightarrow x^2 + 2x - 5 = 0 \Rightarrow p = 2, q = -5, r = 2q - p = -12$
- Q2 (D)** base 1: $x = 1, 4$; index 0: $x = 4, 5$; base -1 : $x = 2, 3$ (index 6, 2 even). $S = 15 = a_4 \Rightarrow \log_2 16 + 4 = 8$
- Q3 (A)** $t = \log_{\sqrt{3}+\sqrt{2}} x \Rightarrow -t^2 = -4, t = \pm 2; x = 5 \pm 2\sqrt{6}; 10 + 1 = 11$
- Q4 (C)** $\sqrt{5} - 2 = (\sqrt{5} + 2)^{-1} \Rightarrow \text{LHS} = t - t = 0 \neq 4, \text{ so } N = 0; 0 + 4d = 12 \Rightarrow d = 3$
- Q5 (D)** $(x + 2)^2 = x(3x + 2) \Rightarrow x^2 - x - 2 = 0 \Rightarrow x = 2$ ($x = -1$ rejected); $\frac{1}{4}r^3 = 2 \Rightarrow r = 2$
- Q6 (C)** $S = \frac{2}{1 - 1/\sqrt{2}} = 4 + 2\sqrt{2} \Rightarrow x^2 - 8x + 8 = 0 \Rightarrow p + q = 0$
- Q7 (D)** $t = \log x: t^2 - 2t - 2 = 0, t_1 + t_2 = 2 \Rightarrow P = 10^2; 2, k, 8 \text{ in AP} \Rightarrow k = 5$
- Q8 (B)** $\alpha - \beta = 2\sqrt{2}, \alpha^2 + \alpha\beta + \beta^2 = 4 + 1 = 5 \Rightarrow 10\sqrt{2}; k = 40\sqrt{2} - 10\sqrt{2} = 30\sqrt{2}$
- Q9 (C)** $t_1 t_2 = 2(t_1 + t_2) - 3 \Rightarrow (t_1 - 2)(t_2 - 2) = 1 \Rightarrow t_1 = t_2 = 3 \text{ or } 1; k = 4 \text{ or } 0; \text{ middle term } 2$
- Q10 (A)** $4ar = a + 3ar^2 \Rightarrow 3r^2 - 4r + 1 = 0 \Rightarrow r = \frac{1}{3}; S_\infty = \sqrt{2}/\frac{2}{3} = \frac{3\sqrt{2}}{2}$
- Q11 (B)** $t = 2, \frac{1}{2} \Rightarrow x = 9, \sqrt{3}; \log_3(9\sqrt{3}) = \frac{5}{2}; k = \frac{1}{2}(\frac{5}{2} + 10) = \frac{25}{4}$
- Q12 (D)** $A^2 = A + 2 \Rightarrow A = 2; B^2 = B + 6 \Rightarrow B = 3; G^2 = 6; \log_2 8 = 3$

- Q13 (A) $19 \pm 6\sqrt{2} = (3\sqrt{2} \pm 1)^2 \Rightarrow \alpha - \beta = 2; k = 16/2 = 8$
- Q14 (C) $x^2 = 3(x+6) \Rightarrow (x-6)(x+3) = 0; x > 0 \Rightarrow x = 6; N + x_0 = 1 + 6 = 7$
- Q15 (B) $b = 5, 75 + 2d^2 = 83 \Rightarrow d = 2; a, c = 3, 7; g = \sqrt{21}$
- Q16 (C) $u = \sqrt{x-1}; |u-2| + |u-3| = 1 \Rightarrow 2 \leq u \leq 3 \Rightarrow 5 \leq x \leq 10, 6 \text{ integers}; \frac{2}{3}r^2 = 6 \Rightarrow r = 3$
- Q17 (B) $y = 2^x: y^2 - 6y + 8 = 0 \Rightarrow y = 2, 4 \Rightarrow x = 1, 2; \text{sum } 3 \Rightarrow d = 2; T_5 = 1 + 8 = 9$
- Q18 (A) $\text{product } 1 \Rightarrow G = 1; H = \frac{2 \cdot 1}{6} = \frac{1}{3}; G/H = 3; k^2 = 81 \Rightarrow k = 9; \log_3 9 = 2$
- Q19 (D) $\alpha\beta = 2, (\alpha - \beta)^2 = 16 - 8 = 8; 1, 3, k \text{ in AP} \Rightarrow k = 5$
- Q20 (A) $x = \log_2 64 = 6; \sqrt{49} = 7; y^2 = 6 \cdot 7 = 42$
- Q21 13 $\log 18^{15} = 15(1.2552) = 18.828 \Rightarrow d = 19; \log 18^{-15} = \overline{19}.172 \Rightarrow z = 18; 1 + 3(n-1) = 37 \Rightarrow n = 13$
- Q22 7 $\alpha - \beta = 2\sqrt{5}, \alpha^2 + \alpha\beta + \beta^2 = 36 - 4 = 32 \Rightarrow m = 64 = 2^6 \Rightarrow n = 7$
- Q23 2 $a(1-r+r^2) = 7, a(1+r+r^2) = 13 \Rightarrow b = ar = 3; a+c = 10, ac = 9 \Rightarrow M = 9; \log_3 9 = 2$
- Q24 3 $\text{with } x = \log 2, y = \log 3: \text{numerator and denominator both } 6x^2 + 5xy + y^2 \Rightarrow m = 1; 3(m+d) = 12 \Rightarrow d = 3$
- Q25 23 $4d = 4\sqrt{2} \Rightarrow d = \sqrt{2}, a = -\sqrt{2}; T_{15} = 13\sqrt{2}; q = \sqrt{2} \cdot 5\sqrt{2} = 10; 13 + 10 = 23$

VERIFICATION METHOD

1. **Brute force from the statement** – every equation solved by numerical scanning, integer conditions by exhaustive search; exactly one option must match.
2. **Exact arithmetic along the hand method** – fractions and exact $a + b\sqrt{n}$ numbers; 60-digit decimals where a logarithm is unavoidable.
3. **Build-time check** – after the option swaps, the keyed option is re-matched to the true value before the paper is typeset.
4. **Independent re-solve** – a separate solver, given only the printed paper, solved all 25 from scratch: all 25 answers matched the key; it found no wrong option, duplicate option, typo or out-of-syllabus method. It flagged the page footer touching the inner border, which has been fixed.